

REVIEW

Open Access



Advances in forecasting realized volatility: a review of methodologies

Radmir Mischelevich Leushuis^{1*}  and Nicolai Petkov¹

*Correspondence:
rmleushuis@gmail.com

¹ Bernoulli Institute, University
of Groningen, Nijenborgh 9, 9747
AG Groningen, The Netherlands

Abstract

Over the past decade, volatility modeling has gained increasing importance in quantitative finance, significantly influencing risk management, investment strategies, and policymaking. Traditionally, academic research has emphasized linear models for forecasting realized volatility. However, advances in computational power have enabled the development of more sophisticated approaches, including machine and deep learning models. Despite these advancements, no comprehensive survey currently compares traditional linear methods with newer models in the context of realized volatility forecasting. This survey addresses that gap by analyzing all models used to forecast realized volatility in academic literature from 2000 to the first half of 2024. It highlights key academic contributions and examines the empirical characteristics of realized volatility. Our findings indicate that the top-performing hybrid convolutional neural network and long short-term memory model surpasses other models in forecasting accuracy. These results highlight the ability of such models to capture a broader range of factors driving realized volatility, affirming their valuable role in quantitative modeling.

Keywords: Realized volatility forecasting, Time series analysis, Linear models, Machine learning, Deep learning, Financial time series

JEL Classification: C45, C53, C58, G12, G17

Introduction to stock market volatility

Over the past century, advances in digitalization, information processing, and financial modeling have transformed the financial industry from a network of small local banks into a complex global system with significant influence on the world economy (Liermann and Stegmann, 2019, p. 1). At the core of this system are the stock and derivatives markets, with the latter having an estimated notional value nearly 10 times greater than the global gross domestic product (Maverick 2024).

Given the massive volume of capital traded in the derivatives market, it is critical for financial institutions (e.g., hedge funds, trading firms, insurers, and banks) to accurately price the derivatives they trade. A key variable in pricing these instruments is the volatility of the underlying asset. Volatility reflects the distribution of an asset's returns and captures the degree of price fluctuation over time. Higher volatility signals greater

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

uncertainty and risk, as prices are more likely to experience significant swings. Consequently, volatility plays a crucial role in risk management.

Beyond derivative pricing and risk assessment, traders can directly profit from volatility itself, provided they possess more accurate forecasts than market makers. This is particularly relevant in options trading, a common form of derivative. Market makers use volatility forecasts to price options, which allows others to infer the market's expectation of future (i.e., implied) volatility from the quoted option price. If an investor's own forecast of future volatility exceeds the implied volatility, they should buy the option. If their forecast is correct (i.e., the realized or actual volatility over a period is higher than the implied volatility), the option will have been underpriced, resulting in a profit.

Accurate volatility forecasting is therefore essential in modern finance. Improving forecast accuracy depends on effectively modeling the underlying volatility dynamics. This survey reviews and compares all models used in the academic literature to forecast realized volatility from 2000 through the first half of 2024. It is the first comprehensive review to evaluate traditional linear models alongside more advanced approaches, including convolutional neural networks (CNNs), long short-term memory (LSTM) models, heterogeneous autoregressive neural networks (HARNets), and transformers. The survey examines the design and empirical performance of each model using standard goodness-of-fit metrics. It also identifies research gaps and recommends directions for enhancing volatility forecasting techniques. Throughout the analysis, we address two central questions:

1. Which forecasting models are most valuable to academic researchers forecasting realized volatility, and how do they differ in their assumptions and designs?
2. What key empirical characteristics of realized volatility do these models address and how effectively do they capture these features?

The remainder of Section “[Introduction to Stock Market Volatility](#)” discusses the theoretical foundations of (realized) volatility, outlines its empirical properties, and describes the methodology of this survey and related work. Section “[Analysis of applied forecasting models](#)” reviews the academic literature on volatility forecasting models. Section “[Conclusion](#)” summarizes the findings, identifies gaps in current knowledge, and suggests future research directions to improve forecasting performance.

Theoretical volatility—square root of integrated variance

We begin with theoretical volatility, which refers to the continuous and instantaneous price fluctuations described by financial theory. This concept is closely linked to Brownian motion, a fundamental physical science idea applied to financial time series to model asset price movements as random paths. Initially developed to describe bond price behavior, this form of Brownian motion is characterized by independent, stationary, and normally distributed increments (Bachelier 1900). However, applying this model directly to asset prices is problematic because it allows for negative values, whereas asset prices are strictly non-negative. To resolve this, Samuelson (1965) proposed modeling the returns of asset prices, rather than the prices themselves. This

transforms the asset's price path into an exponential value, meaning that at time t , S_t , follows a log-normal diffusion process represented by

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t. \quad (1)$$

Here, dW_t denotes the Brownian motion process, which captures the inherent randomness of asset movement. μ represents the expected instantaneous return, adjusted for dividend yield, and σ denotes the instantaneous return volatility. Using Ito's lemma, the change in the logarithm¹ of the asset price, S_t , is given by

$$d \log(S_t) = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t. \quad (2)$$

The expected return, μ , may also be time-dependent, denoted μ_t , without loss of generality. σ again represents volatility, and σ^2 denotes variance. A key assumption in this model is that volatility σ remains constant over time. However, this assumption often fails in practice. Empirical studies of historical one-year volatilities for stock indices reveal significant time variation in volatility levels (Javaheri, 2005, p. 2), prompting the development of models that incorporate dynamic and time-varying volatility.

One such class is stochastic volatility (SV) models, which treat volatility (σ_t) as a random and time-varying variable. These models offer a flexible framework capable of incorporating empirically observed features of volatility, such as clustering and fat tails (Meddahi 2003). Deterministic volatility σ is a special, degenerate case within this broader SV framework (Javaheri, 2005, p. 20). SV models describe the stochastic process for variance (squared volatility, σ_t^2) as

$$d\sigma_t^2 = \alpha(\sigma_t^2)dt + \beta(\sigma_t^2)dZ_t. \quad (3)$$

Here, dZ_t is a Brownian motion independent of the previously defined dW_t . This formulation allows Eqs. (1) and (2) to reflect time-varying volatility σ_t instead of the constant σ . Functions $\alpha(\sigma_t^2)$ and $\beta(\sigma_t^2)$ are typically assumed to satisfy the following conditions:

1. Both exhibit locally bounded path variation over interval $[0, \infty)$.
2. Both satisfy Lipschitz continuity.

These relatively mild assumptions still permit volatility σ_t to exhibit jumps, long memory, and non-stationarity without requiring an unconditional mean (Barndorff-Nielsen and Shephard 2002). Notably, this flexible formulation does not demand detailed prior knowledge of the stochastic processes governing $\alpha(\sigma_t^2)$ and $\beta(\sigma_t^2)$. Instead, they can be treated as bounded and Riemann integrable. Under these assumptions, σ_t^2 is considered a semimartingale with a continuous local martingale component (Barndorff-Nielsen and Shephard 2002). The presented SV framework allows for deeper analysis

¹ In financial calculations, logarithms are used as an approximation for numerical stability and tractability and are based on the Taylor Series expansion around $x = 1$, where $\log(x) \approx x - 1$. For returns, this means $\log\left(\frac{S_t}{S_{t-1}}\right) \approx \frac{S_t}{S_{t-1}} - 1$, which simplifies computations of percentage returns while still being accurate for small returns.

and interpretation of the structure of variance and volatility. A central concept within this approach is the integrated variance $\sigma_{0,T}^2$, defined as

$$\sigma_{0,T}^2 = \int_0^T \sigma_t^2 dt. \quad (4)$$

This measure represents the cumulative variance accumulated up to time T for process σ_t^2 . It allows for the incorporation of various features of volatility, including sudden jumps and long-term dependencies, providing a comprehensive view of how volatility evolves over time. As such, integrated variance is a key metric enabling the practical use of SV models.

Realized volatility

Although theoretically important, the integrated variance $\sigma_{0,T}^2$ cannot be directly observed in practice. This limitation arises because it is a continuous-time measure that would require information about the asset's instantaneous volatility at every moment. In reality, only discrete and finite market data points are available. The resolution of real-world financial data is constrained by both trading hours and transaction frequency, making it impossible to capture every instantaneous volatility fluctuation.

This lack of continuous data renders the exact calculation of integrated variance, and thus volatility, impractical. Therefore, accurate approximations are necessary. The most widely used approximation is the realized variance, which is computed from high-frequency return data. Unlike theoretical measures, realized variance is observable and captures actual market behavior, making it more suitable for empirical financial modeling. It serves as a bridge between continuous-time theoretical constructs and discrete-time real-world observations. Mathematically, the realized variance $\tilde{\sigma}_t^2$ at time t over a fixed interval divided into N subintervals is given by

$$\tilde{\sigma}_t^2 = \sum_{n=1}^N r_{t+\frac{n}{N}}^2, \quad (5)$$

where $r_{t+\frac{n}{N}}$ represents the log-return of the asset from time t to $t + \frac{n}{N}$, with the fixed interval typically chosen between 1 and 5 min. The log-return at time t is defined as

$$r_t = \log \left(\frac{S_t}{S_{t-1}} \right) = \log(S_t) - \log(S_{t-1}), \quad (6)$$

where S_t denotes the asset's price at time t . The realized variance aggregates squared returns over these small intervals, providing an empirical estimate of the integrated variance, $\sigma_{0,T}^2$. Because it is observable, realized variance enables the application of quantitative models without treating volatility as a latent variable.

Importantly, realized variance $\tilde{\sigma}_t^2$ is a robust and consistent estimator of integrated variance, as it asymptotically converges to $\sigma_{0,T}^2$. This convergence is demonstrated in Barndorff-Nielsen and Shephard (2002). As the expected value of realized variance approaches the true integrated variance, realized volatility likewise converges to true volatility. With this theoretical groundwork in place, we now turn to the empirical

characteristics of realized volatility, which offer valuable insight into actual market behavior.

Properties of volatility

This section outlines the empirical characteristics of realized volatility as identified in academic research. These characteristics help clarify the assumptions underlying the models discussed in the next section and serve as benchmarks for developing new models that more accurately reflect market behavior.

Financial econometrics, risk management, and asset pricing literature highlight six key empirical characteristics of volatility:

- C1 Autocorrelation (volatility clustering): Volatility tends to show autocorrelation and clustering over time (Ding et al. 1993). This means current realized volatility is correlated with its own past values. High-volatility periods are typically followed by more high-volatility periods, and similarly for low-volatility periods (Ding et al. 1993).
- C2 Asymmetry (skewness): Volatility behaves asymmetrically in different market conditions. Specifically, it tends to rise more sharply during downturns than it falls during upturns (Dzieliński et al. 2018). This leverage effect implies that negative returns increase future volatility more than positive returns of the same magnitude.
- C3 Leptokurtosis (fat tails): Volatility distributions often display leptokurtosis, featuring fatter tails and sharper peaks than a normal distribution (Cheng and Hung 2011). This increases the probability of extreme volatility events. The phenomenon is closely tied to price *jumps*, where sudden news or shocks cause large movements that amplify these fat tails.
- C4 Mean reversion (baseline level): Despite short-term clustering and autocorrelation (C1), volatility exhibits mean reversion over longer periods. This means it tends to fluctuate around a long-term average or baseline level (Bali and Demirtas 2008).
- C5 Seasonality (periodic cycles): Volatility displays predictable seasonal patterns driven by cyclical factors such as earnings announcements, holidays, or fiscal reporting dates. As a result, some assets experience regular shifts in volatility during specific times of the year (Richter and Sørensen 2002).
- C6 Co-movement (cross-asset correlation): Asset returns—and hence their volatilities—often move in tandem with other assets (Baker and Wurgler 2012). This co-movement arises during shared macroeconomic shocks or systemic market events and highlights the interconnectedness of financial instruments.

With these characteristics in mind, the following subsection explores how researchers have applied them in forecasting models. We review key modeling approaches and their theoretical underpinnings drawn from the academic literature.

Survey method and related work

To conduct this survey, we identified key models used for forecasting realized volatility by reviewing academic articles tagged with terms related to *forecasting realized volatility*. Our dataset includes publicly available papers published between 2000 and

the first half of 2024. We focused exclusively on studies that model realized volatility directly, deliberately excluding those that rely on latent volatility assumptions, such as generalized autoregressive conditional heteroskedasticity (GARCH) models.

This approach led us to identify 32 core models from 41 influential articles. We classify these models into three main categories: linear, machine learning (ML), and deep learning (DL) models. Each category encompasses a variety of approaches, many of which are extensions designed to address specific characteristics of realized volatility. This survey discusses them in detail to answer two primary research questions:

- Q1. Which forecasting models are most valuable to academic researchers forecasting realized volatility, and how do they differ in terms of their assumptions and design?
- Q2. What key empirical characteristics of realized volatility do these models address and how effectively do they capture these features?

Several prior studies (e.g., Kambouroudis et al. 2021; Wang et al. 2016; Wong et al. 2016; Bucci 2017) have compared different forecasting models. However, most focused solely on linear models. Other studies introduced a single new ML or DL model and compared it only with a basic linear benchmark, without offering a broad cross-class evaluation. This survey addresses that gap by providing a systematic comparison across all model classes covered in academic literature, including both traditional linear approaches and advanced ML and DL architectures such as CNN-LSTM, HARNets, and transformers. We assess each model's strengths and limitations and analyze how well they capture the key characteristics of realized volatility. This review is intended to present a complete picture of how various models perform across different empirical settings.

Analysis of applied forecasting models

Next, we examine 32 linear, ML, and DL models that have been employed in publicly available academic research to forecast realized volatility. For each, we describe the model and the specific volatility characteristics it is designed to capture. Table 1 provides an overview of the models, listing the key articles where each is applied. In these studies, model performance is typically assessed using evaluation metrics such as mean squared error (MSE), root mean squared error (RMSE), R^2 , quantile loss function (*QLIKE*), or empirical value-at-risk (VaR) forecasts.

Linear models

The first class of models (linear) is the oldest and most widely used in academic literature for forecasting realized volatility. Within this class, the two primary types are autoregressive moving average (ARMA) and heterogeneous autoregressive (HAR) models.

ARMA models

ARMA employs past time series values (autoregressive) to account for autocorrelations (C1) and past forecast errors (moving average) to account for mean reversion (C4). Thus, it models weakly stationary time series using two polynomials. The standard ARMA(p, q) formulation for realized volatility $\tilde{\sigma}_t$ is

Table 1 Models are used to forecast realized volatility in publicly available academic literature—Corresponding key articles and characteristics

Class	Type	Model	Satisfied Characteristics	Key Articles
<i>Linear</i>	ARMA-based	ARMA	C1, C4	Pong et al. (2004)
				Kambouroudis et al. (2016)
		ARIMA	C1, C4	Kambouroudis et al. (2016)
		log-ARIMA	C1, C3, C4	Bolko (2020)
		ARFIMA	C1, C4	Pong et al. (2004)
				Koopman et al. (2005)
				Ishida and Watanabe (2009)
				Kambouroudis et al. (2016)
		log-ARFIMA	C1, C3, C4	Pong et al. (2004)
				Koopman et al. (2005)
				Ishida and Watanabe (2009)
		IV-ARFIMA	C1, C4	Chen et al. (2010)
				Kambouroudis et al. (2016)
				Bazhenov and Fantazzini (2019)
		asym. ARFIMA	C1, C2, C4	Degiannakis (2008)
				Kambouroudis et al. (2016)
		VAR	C1, C4, C6	Andersen et al. (2003)
		VARFIMA	C1, C4, C6	Andersen et al. (2003)
				Chen et al. (2010)
	HAR-based	HAR	C1, C4	Corsi (2009)
		log-HAR	C1, C3, C4	Corsi et al. (2012)
		T-HAR	C1, C4, C5	Wang et al. (2016)
		J-HAR	C1, C3+, C4	Andersen et al. (2007)
				Patton and Sheppard (2015)
				Kambouroudis et al. (2021)
		A-HAR	C1, C2, C4	Patton and Sheppard (2015)
				Gong and Lin (2019)
				Kambouroudis et al. (2021)
		JA-HAR	C1, C2, C3+, C4	Patton and Sheppard (2015)
				Gong and Lin (2019)
				Kambouroudis et al. (2021)
		3-HAR	C1, C4	Gong and Lin (2019)
		IV-HAR	C1, C4	Kambouroudis et al. (2021)
		GHAR	C1, C4, C6	Čech and Barunik (2017)
<i>ML</i>	Regularized Regr.	LASSO	C1, C4, C5	Audrino and Knaus (2016)
				Ding et al. (2021)
				Christensen et al. (2023)
		Ridge	C1, C4, C5	Wei et al. (2020)
				Christensen et al. (2023)
				Branco et al. (2024)
	Kernel-based	Elastic Net	C1, C4, C5	Christensen et al. (2023)
		Kernel Regr.	C1+, C4, C5+	Yang et al. (2015)
				LeBaron (2018)
				Şanlı et al. (2023)
	Tree-based	RF	C1+, C3, C2, C4+, C5, C6	Luong and Dokuchaev (2018)
				Gupta and Pierdzioch (2022)
				Demirer et al. (2024)
				Bouri et al. (2021)

Table 1 (continued)

Class	Type	Model	Satisfied Characteristics	Key Articles
DL	CNN-based	XGBoost	C1+, C2, C3, C4+, C5+, C6	Wu and Wang (2021) Teller et al. (2022)
		XGBoost-ARIMA	C1+, C2, C3, C4+, C5+, C6	Wang and Guo (2020)
		LightGBM	C1+, C2, C3, C4+, C5+, C6	Wu and Wang (2021)
		LightGBM-NN	C1+, C2+, C3+, C4+, C5+, C6+	Zhang (2022)
		CNN	C1+, C2+, C3+, C4+, C5+, C6+	Betz et al. (2023)
		HARNet	C1+, C2+, C3+, C4+, C5+, C6+	Reisenhofer et al. (2022)
		LSTM-based	LSTM	Bucci (2020)
				Souto and Moradi (2023)
		LSTM-CNN	C1+, C2+, C3+, C4+, C5+, C6+	Vidal and Kristjanpoller (2020)
		Transformers	C1+, C2+, C3+, C4+, C5+, C6+	Ma et al. (2024) Ramos-Pérez et al. (2021) Frank (2023)

+ Indicates a higher capability to model the characteristics associated with the feature marked

$$\tilde{\sigma}_t = c + \sum_{i=1}^p \alpha_i \tilde{\sigma}_{t-i} + \sum_{j=1}^q \beta_j \epsilon_{t-j} + \epsilon_t, \quad (7)$$

where p and q denote the autoregressive and moving average orders, respectively, ϵ_t is the error term at times t and c , α_i for $i = 1, \dots, p$, and β_j for $j = 1, \dots, q$ are the estimated coefficients. The error term is assumed to be normally distributed with mean zero and variance E (i.e., $\epsilon_t \sim N(0, E)$).

Empirical studies have shown that ARMA models outperform traditional latent-volatility models such as GARCH in terms of fit, as measured by R^2 (Andersen et al. 2003). In a comparison of ARMA, implied volatility,² and GARCH(1,1) models for forecasting exchange rate volatility (1994–1998), Pong et al. (2004) found ARMA (and ARFIMA) models performed best at short horizons using MSE , whereas implied volatility was superior for longer horizons. GARCH forecasts were generally the least accurate but added marginal value for short-horizon implied volatilities. Similarly, Kambouroudis et al. (2016) found that ARMA models delivered the highest accuracy among the three, using MAE , $RMSE$, and adjusted R^2 on data from the S&P 500, Dow Jones Industrial Average, Nasdaq 100, and their respective implied volatility indices.

A key assumption of the ARMA model is the weak stationarity of $\tilde{\sigma}_t$, which does not hold for all assets (Mills and Mills 2015). Weak stationarity requires that the series have constant mean and variance over time and that covariances depend only on the time lag, not the specific time. However, many financial series are nonstationary (Cavaliere and Taylor 2007), violating this assumption. To address this, the autoregressive integrated moving average (ARIMA) model introduces differencing to model the d^{th} order change in volatility instead of volatility itself. This stabilizes the mean by removing trends. The differencing operator, Δ^d , is defined as

² The implied volatility represents the market's expectation of future realized volatility and can be calculated from observed option market prices.

$$\Delta^d \tilde{\sigma} = \begin{cases} \tilde{\sigma}_t - \tilde{\sigma}_{t-1} & \text{if } d = 1 \\ \Delta(\Delta^{d-1} \tilde{\sigma}_t) & \text{if } d > 1 \end{cases} \quad (8)$$

where $d \in \mathbb{Z}$. The ARIMA(p, d, q) model can then be expressed as

$$\Delta^d \tilde{\sigma}_t = c + \sum_{i=1}^p \alpha_i \Delta^d \tilde{\sigma}_{t-i} + \sum_{j=1}^q \beta_j \epsilon_{t-j} + \epsilon_t. \quad (9)$$

The differencing order, d , is typically determined by applying Dickey–Fuller tests until the null hypothesis of a unit root is rejected (Dickey and Fuller 1979). Orders p and q are selected by minimizing the Akaike information criterion (AIC) or Bayesian information criterion (BIC), both of which penalize model complexity (Bolko 2020; Schwarz 1978).

The (log-)ARIMA specification, like the ARMA model, offers a simple way to approximate long-memory behavior in realized volatility (Bolko 2020). Bolko (2020) showed that for one-day-ahead forecasts of log realized volatility across 29 global stock indices, the ARIMA model performs comparably to benchmark models³ in terms of MSE and *QLIKE*. Additionally, Kambouroudis et al. (2016) found that the ARIMA model can outperform more complex models such as autoregressive fractionally integrated moving average (ARFIMA) and GARCH for specific indices, including the NASDAQ100.

One limitation of the ARIMA model is its inability to capture long-term dependencies, which are prevalent in volatility series (C4). This limitation is addressed by the ARFIMA model of Granger and Joyeux (1980), which uses fractional differencing to account for long memory. The model retains the ARIMA (9) structure but redefines Δ^d as

$$\Delta^d \tilde{\sigma}_t = \sum_{k=0}^{\infty} (-1)^k \binom{d}{k} \tilde{\sigma}_{t-k}. \quad (10)$$

Here, $d \in \mathbb{N}$ is fractional. When d is an integer ($d \in \mathbb{Z}$), the ARFIMA model reduces to ARIMA. The infinite sum captures long-term dependencies, making ARFIMA especially effective for modeling mean-reverting behavior (C4).⁴

Empirical results from Pong et al. (2004) indicate that ARFIMA models outperform implied volatility models in R^2 , particularly for pound and yen volatility forecasts at 1-day and 1-week horizons. Ishida and Watanabe (2009) also demonstrated strong performance from ARFIMA models augmented with GARCH(1,1) error terms when forecasting the Nikkei 225's volatility. These models achieved an R^2 near 60% for one-step-ahead forecasts, although specification tests indicated some remaining model deficiencies.

A well-established ARFIMA variation involves modeling log-volatility instead of raw volatility. This has theoretical advantages as variance is additive, and volatility is not. Modeling log-volatility allows for alignment with the additive properties of log-variance.⁵ This transformation enhances interpretability and helps address leptokurtosis (C3) by reducing the influence of outliers. The log-ARFIMA model becomes

³ HAR and ARFIMA models.

⁴ Note that if $|d| < 0.5$, the ARFIMA model is both stationary and invertible (Kambouroudis et al. 2016). When $d \in (0, 0.5)$, the process displays long memory characteristics, while for $d \in (-0.5, 0)$, the process is characterized by antipersistence.

⁵ Aside from a constant, which is absorbed into the model's coefficients: $\alpha_1 \log(\tilde{\sigma}_t^2) = 2\alpha_1 \log(\tilde{\sigma}_t) = \alpha_1^* \log(\tilde{\sigma}_t)$.

$$\Delta^d \log(\tilde{\sigma}_t) = c + \sum_{i=1}^p \alpha_i \Delta^d \log(\tilde{\sigma}_{t-i}) + \sum_{j=1}^q \beta_j \epsilon_{t-j} + \epsilon_t. \quad (11)$$

Empirical studies confirm the benefits of this transformation. Alizadeh et al. (2002) showed improved RMSE performance in forecasting U.S. dollar exchange rate volatility, and Koopman et al. (2005) reported better MSE and R^2 outcomes for the S&P 500 between October 2001 and November 2003.

Another extension involves including asymmetric realized volatility terms that vary by asset return direction. This asymmetric ARFIMA approach captures differences in volatility during declines versus rallies. Degiannakis (2008) applied such models to CAC40 and DAX30 intraday volatility, successfully capturing several empirical features. Likewise, Kambouroudis et al. (2016) found these models minimized forecast error (i.e., MAE, RMSE, adjusted R^2) for the S&P 500 and DJIA.

Implied volatility can also serve as a regressor. Kambouroudis et al. (2016) found that while implied volatility alone may underperform, its inclusion improves model accuracy. Bazhenov and Fantazzini (2019) corroborated this, reporting significant MSE improvements across stocks and models. Similarly, in a multivariate setting, Chen et al. (2010) found that including implied volatility in vector-ARFIMA (VARFIMA) models improved forecast accuracy as measured by MSE and MAE.

The final major extension is the VARFIMA model, a multivariate structure that captures co-movement (C6) among volatilities. As volatility is often correlated across assets due to common macroeconomic shocks (Baker and Wurgler 2012), VARFIMA models are designed to account for this. The VARFIMA formulation extends equation (11) to the multivariate case:

$$\Delta^d \log(\tilde{\Sigma}_t) = c + \sum_{i=1}^p A_i \Delta^d \log(\tilde{\Sigma}_{t-i}) + \sum_{j=1}^q B_j E_{t-j} + E_t, \quad (12)$$

where $\tilde{\Sigma}_t$ is an $N \times 1$ vector of realized volatilities with elements $\tilde{\sigma}_{t,k}$ for assets $k = 1, \dots, N$, and the parameter matrices A_i for $i = 1, \dots, p$ and B_j for $j = 1, \dots, q$ are of dimension $N \times N$.

Empirical results from Chen et al. (2010) indicate that VARFIMA models outperform GARCH(1,1), its extensions such as FIGARCH(1,1), and basic SV models. This underscores the importance of accounting for co-movement in realized volatility forecasting. Furthermore, Andersen et al. (2003) find that the VARFIMA class of models yields significantly more accurate volatility forecasts, as measured by R^2 , compared to conventional GARCH and related models.

HAR model

The second type of linear model, HAR, introduced by Corsi (2009), represents a major advancement in volatility modeling. As noted, financial time series often exhibit features such as long memory (C4) and leptokurtosis (C3) (Cheng and Hung 2011). These features are challenging for traditional models like ARMA or vector-ARMA to capture effectively. Drawing on the heterogeneous market hypothesis of Müller et al. (1997), Corsi (2009) argues that volatility emerges from the actions of different market

participants operating over varied time horizons. His model attempts to incorporate these dynamics to better represent long-memory behavior in realized volatility.

Unlike ARMA or VAR models, the HAR model decomposes realized volatility into additive components attributed to agents operating across different time horizons (Corsi 2009). These agents may include high-frequency traders, hedge funds, or pension funds, each of which rebalances portfolios over distinct intervals. To capture this heterogeneity, Corsi (2009) proposed a linear model aggregating the effects of different time scales into a single volatility estimate.

The HAR model includes three main components: daily (D for lag $t - 1$), weekly (W for lag $t - 5$), and monthly (M for lag $t - 22$) realized volatilities.⁶ These components correspond to standard rebalancing periods and are designed to capture the influence of different time horizons on volatility (Corsi 2009). Notably, the $t - 22$ term introduces a long-memory component (C4)—a key distinction from classical ARFIMA or VARFIMA models, which often require many lags and parameters to achieve similar persistence. The standard HAR model is formulated as follows:

$$\tilde{\sigma}_t = \beta_0 + \beta_D \tilde{\sigma}_{t-1}^{(D)} + \beta_W \tilde{\sigma}_{t-1}^{(W)} + \beta_M \tilde{\sigma}_{t-1}^{(M)} + \varepsilon_t, \quad (13)$$

where $\tilde{\sigma}_{t-1}^{(X)} = \sqrt{\frac{1}{N_X} \sum_{i=1}^{N_X} \tilde{\sigma}_{t-i}^2}$ for $X \in [D, W, M]$, with $N_X \in [1, 5, 22]$, and $\varepsilon_t \sim N(0, E)$.

Like ARFIMA models, the HAR model's linearity allows for analytical tractability and straightforward interpretation. However, it is considerably more efficient: it captures persistence with only three terms, whereas an ARMA model would need up to 22 lags to do so. This efficiency marks a significant step forward in forecasting realized volatility.

Empirical evidence suggests that the HAR model outperforms traditional models such as ARFIMA and exponential GARCH (EGARCH) in forecast accuracy. For example, Xiao et al. (2024) compared ARFIMA, HAR, and EGARCH models in predicting idiosyncratic volatility (IVOL), defined as the realized volatility not explained by market-wide movements. Their findings show that the HAR model delivers superior IVOL forecasts. Additionally, Xiao et al. (2024) report that the HAR model demonstrates higher accuracy and better robustness to microstructure noise⁷ when forecasting US stock volatilities from July 1965 to December 2020.

Similar conclusions are drawn by Corsi et al. (2012) and Chin et al. (2016), who compared HAR-type models with GARCH and FIGARCH frameworks. In their study, Chin et al. (2016) analyzed daily realized volatility of the S&P 500 index from February 2008 to February 2015 and found that HAR-based models yielded superior forecast accuracy, even during the 2008 financial crisis. Likewise, Degiannakis et al. (2022) found that the HAR model outperformed Random Walk and AR models in predicting the realized volatility of agricultural commodities, based on metrics such as MSPE and *QLIKE*.

Izzeldin et al. (2019) reported similar results when comparing HAR and ARFIMA models for U.S. stock volatility across various sectors from 2000 to 2010. They noted

⁶ Note that only trading days, when the exchange is operational, are considered and not calendar days.

⁷ Microstructure noise includes high-frequency distortions in financial data, such as bid-ask bounce, market frictions, and transaction costs (Hansen and Lunde 2006).

that ARFIMA's memory parameter, Δ^d , was sensitive to market conditions and sampling frequency, which reduced its predictive accuracy. In contrast, HAR model forecasts remained stable across forecasting horizons, with minimal shifts in sector rankings.

The results from these studies underscore the robustness of the HAR model, which consistently delivers superior forecast accuracy across different markets and time periods. This performance likely stems from its ability to effectively capture persistent dependencies in realized volatility, something traditional models often struggle with Degiannakis et al. (2022). Nonetheless, academic literature has introduced several extensions to the HAR model that further enhance its predictive capabilities.

Many of these extensions mirror those used in ARFIMA models, demonstrating their effectiveness. First, modeling the logarithm of volatility rather than volatility itself helps address leptokurtosis (C3) and produces more accurate and theoretically sound forecasts. Second, transforming the HAR model into a multivariate framework enables it to capture dependencies and spillovers between assets, improving its ability to represent co-movement (C6) and enhancing predictive performance (Souček and Todorova 2013). This is supported by Čech and Barunik (2017), who introduced the multivariate GHAR model. Their findings show that the model remains robust to heteroskedasticity and nonnormal data and delivers reliable forecasts, as measured by RMSE, even in the presence of noise and non-Gaussian characteristics. The HAR model was further developed in the work of Wang et al. (2016), who relaxed the assumption of constant coefficients and allowed the model parameters to vary over time. This extension is referred to in this survey as the time-varying HAR (T-HAR) model. Wang et al. (2016) argued that realized volatility is subject to structural breaks and shifts in economic regimes due to factors such as business cycles, extreme events, and policy interventions. By incorporating time-varying coefficients, the T-HAR model can better capture the seasonality (periodic cycle) property (C5) of realized volatility. This is supported by the out-of-sample results reported in their study, which show that the T-HAR model delivers more accurate forecasts than both the constant-coefficient HAR model and Markov Regime Switching models, based on MSE-type metrics, for the S&P 500 index from January 1996 to June 2013.

An additional extension to the HAR model is proposed by Andersen et al. (2007), who introduce a jump regressor component. This version, referred to as the jump-HAR (J-HAR) model in this survey, includes a jump regressor defined as the excess realized variance over the realized bi-power variation.⁸ This component helps to capture the leptokurtosis (fat tails) characteristic (C3) of realized volatility. Their findings show that incorporating this jump component significantly improves forecasting accuracy, particularly for long-term monthly forecasts. For the S&P 500 index and T-Bonds, the R^2 values for the J-HAR model were three times higher than those of the traditional HAR model.

Another extension involves using separate HAR regressors for negative and positive returns. This version, referred to as the asymmetric-HAR (A-HAR) model in this survey, addresses the asymmetry property (C2) of realized volatility by accounting for different

⁸ Another consistent estimate for integrated variance, which smooths out jumps.

volatility behavior during market upturns and downturns (Dzieliński et al. 2018). To address this asymmetry, Patton and Sheppard (2015) propose a set of models based on signed realized volatilities. Their results indicate that past negative return volatility has a stronger influence on future volatility than positive return volatility. Moreover, they find that the impact of volatility jumps on future volatility depends on the sign of the jump; negative jumps lead to substantially higher future volatility than positive ones. A simple jump-A-HAR (JA-HAR) model incorporating these findings achieves significantly higher out-of-sample forecast accuracy than traditional HAR models, with forecasting horizons ranging from 1 day to 3 months (Patton and Sheppard 2015).

A distinct adaptation, termed the 3-HAR model in this survey, was introduced by Gong and Lin (2019). They evaluate the original HAR model and various extensions using data on the S&P 500 index from January 1996 to June 2013. Their results indicate that the 3-HAR model, which decomposes realized volatility into three separate linear components (i.e., high-frequency, low-frequency, and trend volatility) outperforms other HAR-type models in predicting future volatility. Like Patton and Sheppard (2015), they find that adding a jump component enhances forecast accuracy, particularly during periods of market stress, such as 1999–2003 and 2008–2013.

Kambouroudis et al. (2021) evaluated several HAR extensions, including the integration of implied volatility (IV) as a regressor. Their findings revealed that IV plays a crucial role in improving forecast accuracy. The resulting IV-HAR model consistently outperforms all other HAR variants without IV across time and markets. However, they also found that no single IV-HAR extension dominates in all markets. For instance, including overnight returns as a regressor enhances performance in most markets, while the inclusion of realized volatility of volatility is only beneficial for two US indices. The inclusion of asymmetry (leverage) effects yields mixed results, improving forecast accuracy in only five markets.

The next subsection introduces ML models, which build on concepts from the linear modeling domain but apply more advanced techniques.

Machine learning models

This section introduces ML techniques, which often extend the ideas underlying linear models (e.g., HAR) but employ more sophisticated quantitative methods. First, we discuss regularized regressions as a direct extension of the HAR framework. Second, we introduce kernel regression, the first non-parametric method analyzed in this survey. Finally, we explore tree-based models.

Regularized regression

A natural extension of the HAR model is the incorporation of regularization terms. Regularized regression models, while still linear, include many dependent variables and apply shrinkage techniques to prevent overfitting. These shrinkage methods reduce the magnitude of coefficients for less relevant variables, effectively eliminating them and allowing the model to include more predictors with a reduced risk of overfitting (Hastie et al., 2009, p. 61). In the context of realized volatility, three primary types of regularized regressions are examined: least absolute shrinkage and selection operator (LASSO),

ridge regression, and elastic net regression. Each method adds a penalty term, denoted as C , to the objective function, defined as

$$\text{LASSO :} \quad C = \lambda \sum_{i=1}^p |\beta_i| \quad (14)$$

$$\text{Ridge :} \quad C = \lambda \sum_{i=1}^p \beta_i^2 \quad (15)$$

$$\text{Elastic Net :} \quad C = \sum_{i=1}^p \lambda_1 |\beta_i| + \lambda_2 \beta_i^2 \quad (16)$$

where λ , λ_1 , and λ_2 are hyperparameters typically optimized via cross-validation, and β_i for $i = 1, \dots, p$ are the linear model parameters. All three methods aim to regularize the regression model, with elastic net serving as a hybrid of LASSO and ridge regression.

In the context of realized volatility forecasting, these regularized models generalize the HAR and ARFIMA frameworks, both of which approximate an $AR(\infty)$ process. Consequently, they are capable of capturing the same fundamental properties of realized volatility as the HAR model (i.e., autocorrelation (C1) and mean reversion (C4)). However, $AR(\infty)$ models are typically approximated by finite-order models such as $AR(k)$ for large k (Audrino and Knaus 2016). Under this approach, regularized regression models retain only the most relevant predictors by shrinking the coefficients of irrelevant variables to zero. If the heterogeneous agent assumption of the HAR model holds, coefficients for lags $k > 22$ would likely be eliminated, and the remaining coefficients would approximate the average realized volatilities used in the HAR model.

Audrino and Knaus (2016) assessed this assumption by applying a LASSO(100) model to the realized volatility of nine assets traded on the New York Stock Exchange (NYSE) from January 2, 2001, to November 15, 2010. Their findings showed that the LASSO(100) and the traditional HAR model exhibit comparable forecast accuracy with both in- and out-of-sample outcomes evaluated using MSPE. They argued that the LASSO's shrinkage property, which selects only the most relevant predictors, helps reduce model dimensionality and improves forecast performance. However, they also found that LASSO is unable to fully replicate the HAR model, suggesting a possible violation of the HAR model's assumed lag structure (1, 5, 22). This result implies that the true volatility dynamics may be more complex than those captured by the HAR framework. Additionally, the LASSO(100) model demonstrates greater robustness to regime shifts than the HAR model, likely due to its ability to incorporate more relevant predictors. This adaptability suggests that regularized regression models may also capture the seasonality characteristic of realized volatility (C5).

Ding et al. (2021) further investigated the performance of the LASSO model relative to the HAR model in forecasting realized volatility. Using daily realized volatility data from eight international stock indices (UK FTSE, Japan N225, US SPX, Germany DAX, China SSEC, India NSEI, Brazil BSVP, and Mexico MXX) spanning 2006 to 2020, they evaluate performance using various forecast metrics, including *MSE*, *QLIKE*, *MCS*, and

VaR. Their coefficient plots indicate that the first 22 lags are the most influential in terms of magnitude and statistical significance, lending some empirical support to the HAR model's lag structure. However, they also note that more distant lags still contain useful information, which contributes to the superior out-of-sample forecast accuracy of the LASSO AR(100) model over the HAR model.

Other studies have also examined the effectiveness of regularization techniques. Wei et al. (2020) analyze the use of ridge regression for forecasting the realized volatility of China's gold futures between March 18, 2011, and June 29, 2018. They find that Ridge regression outperforms traditional HAR-type models based on adjusted R^2 , likely due to its ability to address multicollinearity among lagged regressors, an issue that can negatively impact HAR models (Wei et al. 2020). Similarly, Branco et al. (2024) forecast realized volatility for global stock indices from January 2000 to December 2021. Their results, based on MSE, indicated that both Ridge and LASSO models outperform traditional linear methods.

Despite its potential, the elastic net model remains relatively underexplored in the realized volatility literature. Christensen et al. (2023) compare LASSO, ridge, and elastic net models for realized volatility forecasting. In their study, they extend the regressors beyond lagged volatility to include implied volatility and macroeconomic variables to evaluate the effects of shrinkage. Their results show that while Ridge regression performs slightly worse than LASSO based on MSE, it still outperforms benchmark models, particularly when a larger number of predictors are included.

Kernel regressions

Another method related to the HAR model is kernel regression⁹, a non-parametric technique. Kernel regression fits a simple linear model at each target point and weights observations based on their proximity to that point. This is done using a kernel (weighting) function $K_h(x_0, x_i)$ for $i = 1, \dots, N$, where h is a bandwidth parameter that defines the width of the local neighborhood. The kernel assigns weights to each observation x_i based on its distance to the target point x_0 . These memory-based methods require minimal training because the model essentially consists of the full dataset (Hastie et al., 2009, p. 191).

In its basic form, kernel regression solves a linear weighted least squares problem at each target point x_0 . This involves minimizing the weighted sum of squared differences between observed values y_i and those predicted by the local linear models. The optimization problem can be written as

$$\min_{a_0, b_0} \sum_{i=1}^N K_h(x_0, x_i) (y_i - a_0 - b_0 x_i)^2, \quad (17)$$

where the forecast $\hat{y}_0$ is a linear combination of x_0 , $\hat{a}_0$ and $\hat{b}_0$:

$$\hat{y}_0 = \hat{a}_0 + \hat{b}_0 x_0. \quad (18)$$

⁹ Also known as local linear regression.

Note that by plugging in the least square solution to Eq. (17) into Eq. (18) one can write:

$$\hat{y}_0 = \sum_{i=1}^N l_i(x_0) y_i, \quad (19)$$

where $l_i(x_0)$ is a function of $K_h(x_0, x_i)$, $\hat{a}_0$ and $\hat{b}_0$, making the forecast, $\hat{y}_0$, linear in all y_i 's for $i = 1, \dots, N$. Note that the kernel and the bandwidth parameter h are usually chosen using cross-validation (Hastie et al., 2009, p. 199). To apply kernel regressions to time series forecasting, the framework must be slightly adapted. In the context of realized volatility forecasting, LeBaron (2018) proposes a framework in which the kernel uses the same regressors as the HAR model. This design ensures that the kernel regression model can, at a minimum, capture the same characteristics of realized volatility as the HAR model, including autocorrelation (C1) and mean reversion (C4).

The kernel function assigns weights based on the distance between current and past average realized volatilities. Specifically, at time step t , it weighs $Y_t = [\tilde{\sigma}_{t-1}^{(D)}, \tilde{\sigma}_{t-1}^{(W)}, \tilde{\sigma}_{t-1}^{(M)}]$ relative to past observations Y_{t-i} for $i = 1, \dots, t$. This approach effectively produces a weighted version of the HAR model, in which kernel regression searches for the past regime that most closely matches the present. Owing to its locality property, this framework is particularly suited to capturing regime shifts and seasonal patterns (C5).

Empirical results from LeBaron (2018), who modeled the realized volatility of the Dow Jones Industrial Index from 1933 to 1999, indicate that kernel regressions can capture nonlinear features without requiring complex model specifications. Their results show improved out-of-sample forecast accuracy, based on R^2 , compared with the traditional HAR model. These findings also suggest that kernel regressions can better model nonlinear autocorrelations in realized volatility, even during extreme events such as the Great Depression of the early 1930s (LeBaron 2018).

Kernel regression methods have also been applied by Yang et al. (2015), who examined structural breaks in the realized volatilities of the Shanghai Stock Exchange Composite Index and five sectoral stock indices from January 2000 to December 2011. Their findings confirm that kernel regressions are effective in capturing nonlinear features, which are critical for accurate volatility forecasting. Additionally, their work underscores the importance of accounting for structural breaks in forecasting models.

Şanlı et al. (2023) used kernel regression to forecast Bitcoin's realized volatility from September 2014 to March 2020, a period encompassing the COVID-19 pandemic. They recommended using exponentially weighted moving averages (EWMA) of past realized volatilities to enhance the kernel's locality features, as opposed to the simple averages employed by LeBaron (2018). Their model outperformed traditional alternatives based on the cumulative sum of squared prediction errors, particularly during regime shifts such as those triggered by the pandemic (Şanlı et al. 2023).

Tree-based models

Another effective class of nonlinear models for time series forecasting is tree-based models. Unlike linear models, which require the specification of relationships between regressors and the target variable in advance, tree-based regressions adopt a non-parametric framework (Christensen et al. 2023). These models can naturally capture

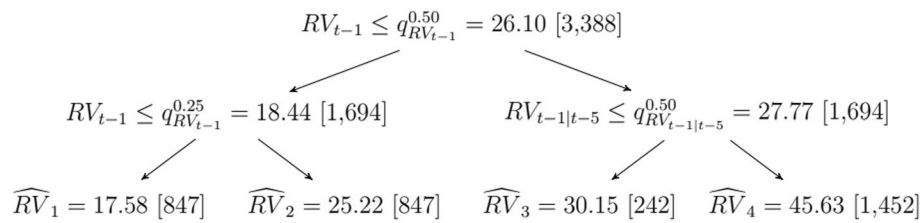


Fig. 1 Basic regression tree for realized volatility forecasting (Christensen et al. (2023)). $q_{t-1}^{0.5}$ represents the median of the realized volatility (RV_t) up to time step $t - 1$, and $q_{t-1|t-5}^{0.5}$ represents the median of the 5-day average realized volatility. The numbers *split [obs]* represent the split point value and the number of observations that fall within this split

nonlinearities and interactions between explanatory variables. The core idea is to recursively partition the feature space into increasingly smaller subspaces using decision trees until a stopping criterion is met. Each terminal node of the tree provides a constant forecast for all observations that fall within that node. This approach is particularly effective at modeling complex data relationships and enables tree-based models to capture various characteristics of realized volatility (C1)–(C6). Owing to their nonlinear structure, these models are especially well-suited for capturing autocorrelation (C1) and mean reversion (C4). A basic example of a regression tree for forecasting realized volatility is illustrated in Fig. 1.

One limitation of tree-based regressions is that they pose a combinatorial optimization problem. There are no closed-form solutions for determining the optimal variable sorting and splitting order (Christensen et al. 2023). Because exhaustively evaluating all possible combinations is computationally intensive, most implementations rely on greedy algorithms that make locally optimal decisions to balance model accuracy and computational efficiency.

A popular and effective approach is the random forest (RF) model, which combines bootstrap aggregation (bagging)¹⁰ with random feature selection for each split (Christensen et al. 2023). By constructing many computationally efficient regression trees and averaging their predictions, the RF model generates a final forecast that improves accuracy and reduces the risk of overfitting. Its key strength lies in its ability to combine numerous inaccurate but uncorrelated forecasts into a single, more accurate prediction.

Empirical results confirm the efficacy of RF models in forecasting realized volatility. For example, Luong and Dokuchaev (2018) studied the realized volatility of the S&P ASX 200 Index from January 2008 to December 2014, a period that includes the global financial crisis. Their findings showed that the RF model outperforms the traditional HAR model in terms of *RMSE* and out-of-bag variance explained. They also demonstrate that incorporating cleaned implied volatilities and indicators of future volatility direction as explanatory variables significantly improves both absolute and relative forecast accuracy, particularly for short-term horizons. Another study on the application of RF models to forecasting realized volatility was conducted by Gupta and Pierdzioch

¹⁰ Bootstrap aggregating (bagging) is an ensemble modeling technique in which multiple models are trained independently on different subsets of the data that are drawn randomly with replacement.

(2022), who used RF to examine the impact of policy and geopolitical risks on oil price volatility. Their findings show that RFs can manage many variables and capture nonlinear relationships, producing more accurate forecasts than traditional methods based on *RMSE* ratios. These results were especially significant at intermediate to long forecast horizons (Gupta and Pierdzioch 2022). Their conclusions are consistent with those of Demirer et al. (2024), who studied US stock market volatility, and Bouri et al. (2021), who examined Bitcoin volatility. Both studies confirmed the strong predictive performance of RFs, as indicated by standard metrics such as *RMSE*.

A recent advancement in tree-based models involves the application of boosting techniques. Unlike RFs, which generate trees independently using random subsets of features, boosted models construct trees sequentially. Each tree in the sequence is trained to correct the errors made by the previous ones. This iterative refinement improves model accuracy with each new tree. Boosting is particularly well suited for modeling complex and nonlinear datasets, offering improved forecasting accuracy based on *RMSE* and *RMSE* ratios relative to traditional RF models (Zhang 2022). In realized volatility forecasting, boosting also better captures long-term patterns such as seasonality (C5) than RFs. Two of the most effective tree-based boosting models used for this purpose are XGBoost, developed by Chen and Guestrin (2016), and LightGBM, developed by Ke et al. (2017).

Empirical evidence has shown the strong performance of tree-based boosting models in practice. Teller et al. (2022) used XGBoost to forecast the realized volatility of Dow Jones stocks from April 2007 to April 2019 and compared its performance to linear HAR models and LSTM networks. Their results show that XGBoost outperformed both models in one-step-ahead forecasts based on *MSE* and *QLIKE*, highlighting its ability to capture nonlinearities in short-term volatility. In another study, Wu and Wang (2021) found that LightGBM outperformed logistic regression, SVM, and XGBoost in intraday realized volatility forecasting based on *MSE*, though XGBoost's accuracy was comparable to that of LightGBM.

Combining boosted tree models with other model classes can further improve out-of-sample forecast accuracy. For example, the DWT-ARIMA-GSXGB model, which integrates XGBoost with ARIMA, performs better than either model alone when forecasting stock price volatility (Wang and Guo 2020). Similarly, Zhang (2022) showed that an ensemble combining LightGBM with neural networks (NNs) outperformed standalone LightGBM. This result supports the universal approximation theorem, which will be discussed in the next subsection. It also anticipates the value of NNs' approximation capabilities, which are examined below.

Deep learning models

This subsection examines models from the DL domain, which have recently emerged as promising tools for time series forecasting. These models are based on NNs and are characterized by their specific architectures. A key advantage of NNs is their foundation in the universal approximation theorem, which states that a NN can approximate any continuous function with arbitrary accuracy (Hornik et al. 1989). This capability allows NNs to model the full range of realized volatility characteristics (C1)–(C6) with

considerable accuracy, giving them an advantage over traditional linear methods and many ML approaches that capture only subsets of these features.

In the time series literature, the two principal NN architectures are CNNs and LSTMs. We begin with CNNs and their applications in realized volatility forecasting, including a review of the CNN-based HARNet model. We then discuss LSTM models, including LSTM–CNN hybrids and transformers, and their effectiveness in volatility prediction.

Convolutional neural networks

CNNs are a type of feed-forward NN that can effectively model complex time dependencies in time series data. They learn features by applying convolutional kernels with shared weights across the input, allowing them to automatically perform feature engineering. These kernels are moved across both time and feature dimensions, producing translation-equivariant output features.

One of the main advantages of CNNs in time series forecasting is their relatively large receptive field.¹¹ This feature becomes particularly valuable for temporal data. Although individual convolutional layers use computationally inexpensive small kernels, stacking layers increases the receptive field, enabling the model to integrate information over extended periods (Goodfellow et al., 2016, p. 332). This makes CNNs highly effective for capturing long-term dependencies in realized volatility data. As additional convolutional layers are added, the CNN can extract features from increasingly wider temporal windows. Consequently, CNNs are well suited for modeling seasonality (C5) and mean reversion (C4), and they outperform traditional recurrent NNs (RNNs) in forecast accuracy (Borovykh et al. 2017).

A second advantage of CNNs is their capacity to identify localized patterns in asset and time dimensions. By applying convolutional kernels to a tensor containing multiple time series, CNNs can capture cross-asset effects over both short and long horizons. This allows them to model the co-movement characteristic (C6) of realized volatility (Borovykh et al. 2017).

A third benefit is the use of nonlinear activation functions in each layer, which enables CNNs to learn complex nonlinear relationships. This makes CNNs well suited to modeling autocorrelation (C1), asymmetry (C2), and leptokurtosis (C3). Unlike linear models, CNNs can capture nonlinear autocorrelations, offering greater flexibility in representing diverse temporal structures (Borovykh et al. 2017).

Lastly, CNNs are computationally efficient relative to many other DL models (Geras et al. 2016). Their design allows them to fit only the kernel parameters and apply the same weights across the input by sliding the kernel. This weight-sharing mechanism drastically reduces the number of model parameters compared to fully connected networks and improves generalization by mitigating overfitting (Goodfellow et al., 2016, p. 333).

Empirical studies have also demonstrated the success of CNNs in time series forecasting. Borovykh et al. (2018) found that CNNs are capable of capturing long-term dependencies and complex structures, and can process wide input ranges with fewer layers and

¹¹ A receptive field is the area of input data that a single CNN filter covers.

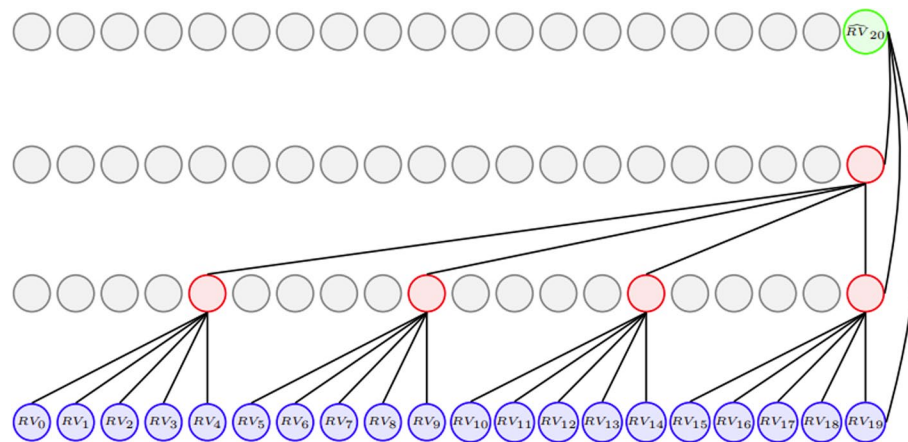


Fig. 2 HARNet framework (Reisenhofer et al. 2022)

parameters than fully connected networks, enhancing out-of-sample accuracy. This is particularly relevant for forecasting realized volatility.

In the context of realized volatility, Betz et al. (2023) showed that CNNs were effective forecasters. They trained CNNs on image-like representations of intraday returns aggregated over the past day, week, and month to predict one-day-ahead realized volatility. Using minute-level data for 28 Dow Jones stocks from 2010 to 2019, their results indicate that CNNs outperformed a benchmark HAR model in both in-sample and out-of-sample RMSE. This suggests that CNNs can detect predictive patterns that HAR models miss, enhancing one-step-ahead forecast accuracy. Another relevant application is the HARNet model developed by Reisenhofer et al. (2022), which will be analyzed in the next subsection.

HARNet

The HARNet model, introduced by Reisenhofer et al. (2022), combines the traditional HAR model with CNNs. HARNet aims to integrate the volatility dynamics assumed in the HAR framework, particularly the persistence of daily realized volatility, with the high forecasting accuracy of CNNs. In its HARNet₂₀ variant, the model uses the past 20 daily realized volatilities to predict the realized volatility for day $t + 21$. This design is inspired by the heterogeneous agent framework of the HAR model, which attempts to capture daily, weekly, and monthly volatility trends. A key innovation of HARNet is its use of dilated convolutional layers, which form a hierarchical structure that exponentially expands the model's receptive field without a corresponding increase in the number of parameters.

In this framework, the initial convolutional layer is crucial for detecting short-term volatility patterns by analyzing recent daily realized volatilities. As the data progresses through subsequent layers, the receptive field grows, enabling the model to learn long-term volatility dynamics. This progression from short- to long-term information is essential for accurately forecasting future volatility (Reisenhofer et al. 2022). Conceptually, the HARNet model can be interpreted as a nonlinear extension of the HAR model, where the contribution (weight) of each daily volatility observation is learned rather than prespecified.

Nonlinearities are incorporated into the HARNet model through the use of a rectified linear unit (ReLU)¹² activation function in each convolutional layer. These functions are essential for enabling the model to capture nonlinear dependencies between past and future volatilities. The combination of nonlinear activation functions with dilated hierarchical convolutional layers makes the HARNet₂₀ model a robust and flexible framework for forecasting realized volatility. The detailed architecture of the HARNet₂₀ model is shown in Fig. 2.

To mitigate common NN issues related to optimization and fitting, Reisenhofer et al. (2022) proposed a novel initialization technique. Their approach ensures that the initial forecasts of the HARNet model replicate those of the baseline HAR model by setting the initial weights of HARNet to the estimated coefficients from the HAR model. This is feasible because the model can be reformulated using convolution operations with average filters, revealing that it is a degenerate case of HARNet. This initialization strategy stabilizes the network's optimization and guarantees that it performs at least as well as the HAR model from the outset. When further optimized, HARNet significantly improves forecast accuracy compared to the baseline HAR model, particularly in terms of the *QLIKE* error¹³ (Reisenhofer et al. 2022). Furthermore, this approach provides a tractable framework for optimizing neural networks, reinforcing its practical value in financial econometrics.

A notable contribution of Reisenhofer et al. (2022) is the demonstration of HARNet's ability to identify the relative importance of different segments of historical data in forecasting future realized volatility. The model emphasizes the realized volatility from the last two days of the previous week as particularly influential. Additionally, an analysis of the second layer's weights in HARNet shows that the influence of weekly data within the past month declines almost linearly with time. This finding indicates that HARNet develops a nuanced understanding of temporal volatility patterns and surpasses the HAR model in weighing past information for volatility forecasting (Reisenhofer et al. 2022).

Long short-term memory networks

Another class of DL models that has proven effective in modeling time series is the LSTM network, introduced by Hochreiter and Schmidhuber (1997). LSTMs are a specialized form of RNN designed to address the limitations of traditional models in capturing long-term dependencies.¹⁴

The primary advantage of LSTMs lies in the architecture of their layers, which consist of memory cells and gating mechanisms. These input, forget, and output gates regulate the flow of information, allowing LSTMs to retain or discard information selectively across time steps. This functionality is critical in time series forecasting, where maintaining long-term context is often essential. For realized volatility forecasting,

¹² Defined as $f(x) = \max(0, x)$. The ReLU adds nonlinearity to the layer by outputting zero for negative values and passing on positive values as they are.

¹³ The *QLIKE* error is computed by comparing the empirical quantiles to the quantiles forecast by the model. Lower *QLIKE* values imply better model performance, as they suggest that the predicted quantiles are closer to the actual quantiles.

¹⁴ Traditional RNNs have difficulties retaining long-term information over multiple time steps and suffer from vanishing gradients during training (Goodfellow et al., 2016, p. 398).

this allows LSTMs to effectively model seasonality (C5) and mean reversion (C4). Furthermore, the gated architecture mitigates vanishing and exploding gradient issues, thereby enabling the network to learn long-term dependencies over extended sequences without common optimization failures (Greff et al. 2016).

Like CNNs, LSTMs employ nonlinear activation functions, enabling them to model complex, nonlinear temporal relationships. This capacity allows LSTMs to capture intricate patterns in realized volatility that linear models typically miss. As a result, LSTMs are well-suited to modeling nonlinear characteristics such as autocorrelation (C1), asymmetry (C2), and leptokurtosis (C3), across both time and asset dimensions. Additionally, LSTMs offer greater robustness to noise and missing data due to their recurrent and gated design, which helps mitigate overfitting and enhances forecast reliability (Hochreiter and Schmidhuber 1997).

Empirical evidence from Bucci (2020) supports the effectiveness of LSTMs in realized volatility forecasting. Their study examined the S&P 500 index from February 1950 to December 2017, comparing several neural network architectures against traditional econometric models like ARFIMA. The results show that LSTMs captured long-term dependencies effectively and outperformed feed-forward, Elman, and Jordan NNs in both MSE and *QLIKE*, especially during high-volatility periods such as the 2008 financial crisis. These findings highlight LSTM's robustness in handling regime shifts. Furthermore, for multistep-ahead forecasts, they outperformed traditional models and matched the performance of other DL approaches in in-sample and out-of-sample contexts (Bucci 2020).

Similar findings were presented by Souto and Moradi (2023), who evaluated LSTM models for forecasting realized volatility during financial turbulence. Their study, which focused on the S&P 500 from November 2017 to November 2022, demonstrates that LSTMs effectively captured elevated volatility levels during market turmoil. This suggests that LSTMs accurately model the leptokurtosis of volatility (C3). When compared to econometric models such as GARCH and HAR, LSTMs yielded more accurate forecasts under volatile conditions, as measured by *RMSE* and *RMSE* ratios (Souto and Moradi 2023).

Further evidence comes from Vidal and Kristjanpoller (2020), who proposed a CNN-LSTM hybrid framework for forecasting gold price volatility. In their architecture, LSTMs captured short-term dependencies while CNNs modeled long-term patterns. Using daily realized volatility data from the London Bullion Market Association from April 1968 to October 2017, and an 80–20% train-test split, their model achieved a 37% reduction in *MSE* compared to the GARCH model and an 18% improvement over standalone LSTMs. The model also outperformed standalone CNNs, traditional NNs, and hybrid GARCH-NN models, highlighting the value of combining LSTM and CNN architectures. However, Ma et al. (2024), in their study of the Chinese SSE index from January 2000 to December 2016, which includes the 2008 global financial crisis and the 2015 Chinese stock market crash, found that CNN-LSTM models outperformed standalone CNNs but not LSTMs, based on out-of-sample *MSE*, *MAE*, and *QLIKE*. Nonetheless, the CNN-LSTM models still outperformed all machine learning methods, including RF, ridge regressions, and SVR.

Transformers

An additional advancement in time series modeling is the transformer architecture, which is a recent evolution of the LSTM framework (Vaswani et al. 2017). The key innovation in transformers is the use of multi-head self-attention mechanisms, allowing the model to *focus* on different segments of the input sequence. This mechanism enhances the model's ability to identify complex patterns and capture long-term dependencies (Vaswani et al. 2017).

Transformers consist of an encoder and a decoder, each composed of multiple layers. The encoder processes the input time series and encodes it into a sequence of vectors. The decoder then uses these encoded representations to generate forecasts for future time steps. Both the encoder and decoder make use of multi-head self-attention layers. A single self-attention layer can be described as follows:

1. **Key, Query and Value Vectors:** For each data point x_t , the transformer layer calculates three vectors:

- Key Vector: $K(x_t) = W^K x_t$
- Query Vector: $Q(x_t) = W^Q x_t$
- Value Vector: $V(x_t) = W^V x_t$

where W^K , W^Q , and W^V are learnable projection matrices. The key vector represents the encoding of the current time step. The query vector allows comparisons between the current time step and all prior time steps. The value vector represents the features of each time step, which are used in forecasting the next time step once the relevant ones are identified. These roles become clearer when steps 2 and 3 are taken into account.

2. **Calculating attention scores:** The attention score $A(x_t, x_j)$ is computed between two data points x_t and x_j for $j < t$ using the query and key vectors. This is done via a dot product followed by a SoftMax function:

$$A(x_t, x_j) = \text{softmax}\left(\frac{Q(x_t) \cdot K(x_j)^T}{d_k}\right) \quad (20)$$

where d_k is a scaling factor equal to the dimensionality of W^K . This factor prevents the dot products from becoming excessively large, which would push the SoftMax function into regions with vanishing gradients. The attention score $A(x_t, x_j)$ quantifies the relevance of time step x_j to time step x_t .

3. **Using Value Vectors:** Once attention scores are computed, the transformer layer uses the value vectors $V(x_j)$ to calculate the output. It aggregates these vectors, weighted by their corresponding attention scores, to produce the output of the self-attention layer:

$$Z_t = \sum_{j < t} A(x_t, x_j) V(x_j). \quad (21)$$

Aggregating over $t = 1, \dots, T$, the resulting vectors can be stacked into a matrix Z .

One can stack N self-attention layers, each with independent random initializations of $[W_p^K, W_p^Q, W_p^V]$ for $p = 1, \dots, N$. Since each layer processes the same input x_i , this results in $Z_1, \dots, Z_N$, which are then combined into a single output Z^* using another learnable projection matrix W^Z . This transforms the self-attention layer into a multi-head self-attention layer, where N denotes the number of *heads*. This design enables the model to simultaneously learn multiple representations of the data in the encoder and decoder layers, thereby enhancing the expressiveness of the model at each time step (Vaswani et al. 2017).

A key advantage of transformers is their ability to detect and model long-term dependencies in time series data. Their self-attention mechanism enables each time step to interact directly with all previous time steps, allowing the model to capture extended temporal correlations (López-Ruiz et al. 2022). This is particularly valuable for realized volatility forecasting, where long-term patterns are important. Another benefit of transformers is their parallel processing capability, in contrast to the sequential nature of LSTMs. This significantly speeds up training and makes transformers more computationally efficient than traditional DL models (Zerveas et al. 2021).

Despite their advantages, transformers have limitations. One major drawback is that their number of parameters grows quadratically with input size, potentially causing computational inefficiencies with very large datasets (Bulatov et al. 2023). Furthermore, transformers require more memory and longer training times than traditional DL models, making them less practical for small datasets. In such cases, CNNs often outperform transformers (Chen et al. 2022).

In the context of realized volatility forecasting, Frank (2023) evaluated temporal fusion transformers (TFTs)—a variant of the transformer architecture—using weekly and monthly realized volatility data from S&P 500 stocks between January 2000 and June 2022. The study found that TFTs, especially when incorporating pooling methods, significantly outperformed other DL and ML models, including LSTMs and Random Forests, based on RMSE and MAE. Moreover, Frank (2023) showed that incorporating sectoral pooling further improved forecast accuracy across all machine learning models.

These findings align with those of Ramos-Pérez et al. (2021), who proposed a multitransformer architecture that combines local and temporal transformers to capture both short- and long-term dependencies. Their study on the S&P 500 from January 2008 to December 2020 compared various neural network architectures. Hybrid models combining transformers and LSTMs outperformed all other approaches based on MAE and RMSE, particularly during high-volatility periods such as the 2020 COVID-19 pandemic.

Bringing together the insights from linear, ML, and DL models, we now turn to the conclusion, which summarizes our key findings and addresses the primary research questions in depth.

Conclusion

This literature survey provides a comprehensive overview of realized volatility forecasting models by addressing two core research questions:

- Q1. Which forecasting models are used in academic research to forecast realized volatility, and how do they differ in terms of their assumptions and design?
- Q2. What key empirical characteristics of realized volatility do these models address and how effectively do they capture these features?

To address these questions, we surveyed publicly available academic articles that referenced the term *forecasting realized volatility*, focusing on studies published between 2000 and H1 2024. Only studies that explicitly modeled realized volatility were included, excluding models where volatility was treated as a latent variable, such as those based on GARCH. Using this selection process, we reviewed the theoretical properties of each identified model. In total, we identified 32 unique models across 41 key articles. Each model incorporates specific empirical features of realized volatility and can be grouped into three broad categories:

1. *Linear models*: Traditional models such as ARIMA and HAR effectively capture basic volatility features like short-term autocorrelation (C1) and mean reversion (C4). Several extensions (e.g., A-HAR and asymmetric ARFIMA (C2); log-ARFIMA, log-HAR, and J-HAR (leptokurtosis (C3)); JA-HAR (asymmetry and leptokurtosis); T-HAR (seasonality (C5)); and VARFIMA and GHAR (co-movement (C6)), address additional characteristics. Nonetheless, these models often fall short in forecasting accuracy due to their limited ability to capture complex nonlinearities.
2. *ML models*: Machine learning methods such as regularized regressions, kernel regressions, and tree-based models offer improved accuracy by capturing nonlinear and complex temporal patterns. Regularized regression and kernel models are particularly strong in modeling short-term dependencies (C1), mean reversion (C4), and seasonality (C5). Tree-based models, especially RF, XGBoost, and LightGBM, further incorporate features like asymmetry (C2), co-movement (C6), and leptokurtosis (C3). These models also exhibit greater resilience to structural shifts in market conditions.
3. *DL models*: CNNs and LSTMs, in particular, demonstrate the highest forecast accuracy. CNN-based models like HARNet retain the beneficial properties of HAR while significantly improving performance. Advanced DL architectures such as Transformers outperform both basic CNNs and LSTMs, effectively addressing the limitations of traditional linear and ML models. Supported by the universal approximation theorem, these models theoretically capture all six empirical characteristics (C1)–(C6) more effectively than other approaches.

Our findings show that DL models generally offer the most accurate forecasts of realized volatility by capturing complex nonlinear patterns and adapting to various market conditions. However, linear models remain important in settings with limited data or where interpretability is paramount. Models like GARCH and ARIMA continue to be widely used in practice due to their simplicity and low data requirements. In contrast, cutting-edge DL models such as Transformers require more memory and training time, as their parameters scale quadratically with input size.

The main theoretical contribution of this survey is that it offers the first direct comparison of traditional linear and ML models with state-of-the-art DL architectures, including CNN-LSTMs, HARNets, and Transformers. Practically, this comparison helps clarify the strengths and limitations of each modeling approach, supporting practitioners in selecting appropriate models based on forecasting goals, interpretability needs, and data availability. Accurate realized volatility forecasts are critical for risk modeling, monetary policy, and regulatory oversight. Our survey also highlights several promising directions for future research, including the development of hybrid models that integrate the strengths of different architectures, and improvements in the computational efficiency and scalability of DL approaches. Progress in these areas will further enhance volatility forecasting and expand its applications in financial risk management and derivative pricing.

Abbreviations

A-HAR	Asymmetric HAR
AIC	Akaike information criterion
ARFIMA	Autoregressive fractionally integrated moving average
ARFIMAX	Autoregressive fractionally integrated moving average with exogenous variables
ARIMA	Autoregressive integrated moving average
ARMA	Autoregressive moving average
BIC	Bayesian information criterion
BSVP	Bovespa index (i.e., a Brazilian stock market index)
CAC	Cotation assistée en continu (i.e., a French stock market index)
CNN	Convolutional neural network
CSI	China Securities Index (i.e., Chinese counterpart of the S&P 500)
DAX	Deutscher Aktienindex
DJIA	Dow Jones Industrial Average
DL	Deep learning
EGARCH	Exponential generalized autoregressive conditional heteroskedasticity
ELBO	Evidence lower bound
EWMA	Exponentially weighted moving average
FIGARCH	Fractionally integrated generalized autoregressive conditional heteroskedasticity
FNN	Feedforward neural network
FTSE	Financial Times Stock Exchange (i.e., a stock index from the UK)
GARCH	Generalized autoregressive conditional heteroskedasticity
GDP	Gross domestic product
HAR	Heterogeneous auto-regressive
IV	Implied volatility
IVOL	Idiosyncratic volatility
J-HAR	HAR model with a jump component
LASSO	Least absolute shrinkage and selection operator
LSTM	Long short term memory
MA	Moving average
MCS	Model confidence set
ML	Machine learning
MSE	Mean-squared error
MXX	Mexico Index (i.e., a Mexican stock market index)
NASDAQ100	National Association of Securities Dealers Automated Quotations (i.e., an US stock market index)
NN	Neural network
NSEI	National Stock Exchange of India
NYSE	New York Stock Exchange
N225	Nikkei 225 (also called the Nikkei Stock Average, a stock market index for the Tokyo stock exchange)
RA	Autoregressive
ReLU	Rectified linear unit
RF	Random forest
RNN	Recurrent neural network
RV	Realized volatility
SPX	Standard and Poor's 500 index (i.e., an US stock market index)
S&P 500	Standard and Poor's 500 Index
SSEC	Shanghai Stock Exchange Composite (i.e., a Chinese stock index)
SV	Stochastic volatility
SVM	Support vector machine
T-bonds	Treasury bonds

TFT	Temporal fusion transformer
T-HAR	Time-varying HAR
UK	United Kingdom
US	United States
VaR	Value at risk
VAR	Vector autoregression
VARFIMA	Vector autoregressive fractionally integrated moving average
VARMA	Vector autoregressive moving average

Acknowledgements

The authors would like to thank the University of Groningen for its support.

Author contributions

RML collected the articles and designed the survey framework, which was used to write the initial draft. NP reviewed and edited the manuscript, advised on changes, and supervised the research. All authors read and approved the final manuscript.

Funding

Not applicable.

Availability of data and materials

Not applicable.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interests.

Received: 13 September 2024 Accepted: 15 October 2025

Published online: 06 January 2026

References

- Alizadeh S, Brandt MW, Diebold FX (2002) Range-based estimation of stochastic volatility models. *J Finance* 57(3):1047–1091
- Andersen TG, Bollerslev T, Diebold FX, Labys P (2003) Modeling and forecasting realized volatility. *Econometrica* 71(2):579–625
- Andersen TG, Bollerslev T, Diebold FX (2007) Roughing it up: including jump components in the measurement, modeling, and forecasting of return volatility. *Rev Econ Stat* 89(4):701–720
- Audrino F, Knaus SD (2016) Lassoing the HAR model: a model selection perspective on realized volatility dynamics. *Econom Rev* 35(8–10):1485–1521
- Bachelier L (1900) Théorie de la spéculation. *Annales Scientifiques de l'École Normale Supérieure* 17
- Baker M, Wurgler J (2012) Comovement and predictability relationships between bonds and the cross-section of stocks. *Rev Asset Pric Stud* 27(3):553–596
- Bali TG, Demirtas KO (2008) Testing mean reversion in financial market volatility: evidence from S & P 500 index futures. *J Futures Mark Futures Opt Deriv Prod* 28(1):1–33
- Barndorff-Nielsen OE, Shephard N (2002) Estimating quadratic variation using realized variance. *J Appl Econom* 17(5):457–477
- Bazhenov T, Fantazzini D (2019) Forecasting realized volatility of Russian stocks using google trends and implied volatility. *Russ J Ind Econ* 12(1):79–88
- Betz N, Heil T, Peter F (2023) The best of both worlds? Predicting realized volatility based on convolutional neural nets and coloured intraday return images. Available at SSRN: <https://ssrn.com/abstract=4584108>
- Bolko AE (2020) Modelling and forecasting fractional volatility. PhD Thesis, School of Business and Social Sciences, Aarhus University. Submitted in partial fulfilment of the requirements for the PhD degree in Economics and Business Economics, August 2020
- Borovykh A, Bohte S, Oosterlee CW (2017) Conditional time series forecasting with convolutional neural networks. arXiv preprint [arXiv:1703.04691](https://arxiv.org/abs/1703.04691)
- Borovykh A, Bohte S, Oosterlee CW (2018) Dilated convolutional neural networks for time series forecasting. *J Comput Finance*. Forthcoming; available at SSRN: <https://ssrn.com/abstract=3272962>
- Bouri E, Gkillas K, Gupta R, Pierdzioch C (2021) Forecasting realized volatility of bitcoin: the role of the trade war. *Comput Econ* 57:29–53
- Branco RR, Rubesam A, Zevallos M (2024) Forecasting realized volatility: does anything beat linear models? *J Empir Finance* 78:101524
- Bucci A (2017) Forecasting realized volatility: a review. *J Adv Stud Finance* 8(16):94–138

- Bucci A (2020) Realized volatility forecasting with neural networks. *J Financ Econom* 18(3):502–531
- Bulatov A, Kuratov Y, Kapushev Y, Burtsev MS (2023) Scaling transformer to 1m tokens and beyond with rmt. *arXiv preprint arXiv:2304.11062*
- Cavaliere G, Taylor AR (2007) Testing for unit roots in time series models with non-stationary volatility. *J Econom* 140(2):919–947
- Čech F, Barunik J (2017) On the modelling and forecasting of multivariate realized volatility: generalized heterogeneous autoregressive (ghar) model. *J Forecast* 36(2):181–206
- Chen R, Kang Z, Zhang L (2010) Forecasting return volatility in the presence of microstructure noise. *Stat Interface* 3(2):145–157
- Chen T, Guestrin C (2016) Xgboost: a scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining, KDD '16, pp 785–794, New York, NY, USA. Association for Computing Machinery
- Chen B, Wang R, Ming D, Feng X (2022) ViT-P: rethinking data-efficient vision transformers from locality. *arXiv preprint arXiv:2203.02358*
- Cheng W-H, Hung J-C (2011) Skewness and leptokurtosis in GARCH-typed VaR estimation of petroleum and metal asset returns. *J Empir Finance* 18(1):160–173
- Chin WC, Lee MC, Yap GLC (2016) Modelling financial market volatility using asymmetric-skewed-ARFIMAX and-HARX models. *Eng Econ* 27(4):373–381
- Christensen K, Sigggaard M, Veliyev B (2023) A machine learning approach to volatility forecasting. *J Financ Econom* 21(5):1680–1727
- Corsi F (2009) A simple approximate long-memory model of realized volatility. *J Financ Econom* 7(2):174–196
- Corsi F, Audrino F, Renò R (2012) HAR modeling for realized volatility forecasting, chapter Fifteen. Wiley, Hoboken, pp 363–382
- Degiannakis S (2008) ARFIMAX and ARFIMAX-TARCH realized volatility modeling. *J Appl Stat* 35(10):1169–1180
- Degiannakis S, Filis G, Klein T, Walther T (2022) Forecasting realized volatility of agricultural commodities. *Int J Forecast* 38(1):74–96
- Demirer R, Gupta R, Pierdzioch C (2024) Realized stock-market volatility: do industry returns have predictive value? Springer, Cham, pp 377–405
- Dickey DA, Fuller WA (1979) Distribution of the estimators for autoregressive time series with a unit root. *J Am Stat Assoc* 74(366):427–431
- Ding Z, Granger CW, Engle RF (1993) A long memory property of stock market returns and a new model. *J Empir Finance* 1(1):83–106
- Ding Y, Kambouroudis D, McMillan DG (2021) Forecasting realised volatility: does the LASSO approach outperform HAR? *J Int Financ Mark Inst Money* 74:101386
- Dzieliński M, Rieger MO, Talpsepp T (2018) Asymmetric attention and volatility asymmetry. *J Empir Finance* 45:59–67
- Frank J (2023) Forecasting realized volatility in turbulent times using temporal fusion transformers. Technical report, FAU Discussion Papers in Economics
- Geras KJ, Mohamed A, Caruana R, Urban G, Wang S, Aslan Ö, Philipose M, Richardson M, Sutton C (2016) Blending LSTMs into CNNs. Presented at the 4th international conference on learning representations, ICLR Workshop
- Gong X, Lin B (2019) Modeling stock market volatility using new HAR-type models. *Physics A* 516:194–211
- Goodfellow I, Bengio Y, Courville A (2016) Deep learning. MIT Press, Cambridge
- Granger CWJ, Joyeux R (1980) An introduction to long-memory time series models and fractional differencing. *J Time Ser Anal* 1(1):101–121
- Greff K, Srivastava RK, Koutník J, Steunebrink BR, Schmidhuber J (2016) LSTM: a search space odyssey. *IEEE Trans Neural Netw Learn Syst* 28(10):2222–2232
- Gupta R, Pierdzioch C (2022) Forecasting the realized variance of oil-price returns: a disaggregated analysis of the role of uncertainty and geopolitical risk. *Environ Sci Pollut Res* 29:52070–52082
- Hansen PR, Lunde A (2006) Realized variance and market microstructure noise. *J Bus Econ Stat* 24(2):127–161
- Hastie T, Tibshirani R, Friedman J (2009) The elements of statistical learning: data mining, inference, and prediction. Springer series in statistics, 2nd edn. Springer, Berlin
- Hochreiter S, Schmidhuber J (1997) Long short-term memory. *Neural Comput* 9(8):1735–1780
- Hornik K, Stinchcombe M, White H (1989) Multilayer feedforward networks are universal approximators. *Neural Netw* 2(5):359–366
- Ishida I, Watanabe T et al (2009) Modeling and forecasting the volatility of the Nikkei 225 realized volatility using the ARFIMA-GARCH model. Hitotsubashi University, Institute of Economic Research
- Izzeldin M, Hassan MK, Pappas V, Tsionas M (2019) Forecasting realised volatility using ARFIMA and HAR models. *Quant Finance* 19(10):1627–1638
- Javaheri A (2005) Inside volatility arbitrage: the secrets of skewness, 1st edn. Wiley, Hoboken
- Kambouroudis DS, McMillan DG, Tsakou K (2016) Forecasting stock return volatility: a comparison of GARCH, implied volatility, and realized volatility models. *J Future Mark* 36(12):1127–1163
- Kambouroudis DS, McMillan DG, Tsakou K (2021) Forecasting realized volatility: the role of implied volatility, leverage effect, overnight returns, and volatility of realized volatility. *J Future Mark* 41(10):1618–1639
- Ke G, Meng Q, Finley T, Wang T, Chen W, Ma W, Ye Q, Liu T-Y (2017) Lightgbm: a highly efficient gradient boosting decision tree. In: Advances in neural information processing systems, 30
- Koopman SJ, Jungbacker B, Hol E (2005) Forecasting daily variability of the S & P 100 stock index using historical, realised and implied volatility measurements. *J Empir Finance* 12(3):445–475
- LeBaron BD (2018) Forecasting realized volatility with kernel ridge regression. Available at SSRN: <https://ssrn.com/abstract=3229272>
- Liermann V, Stegmann C (2019) The impact of digital transformation and FinTech on the finance professional. Palgrave Macmillan, Basingstoke

- López-Ruiz S, Hernández-Castellanos C I, Rodríguez-Vázquez K (2022) Multi-objective framework for quantile forecasting in financial time series using transformers. In: Proceedings of the genetic and evolutionary computation conference, pp 395–403
- Luong C, Dokuchaev N (2018) Forecasting of realised volatility with the random forests algorithm. *J Risk Financ Manag* 11(4):61
- Ma W, Hong Y, Song Y (2024) On stock volatility forecasting under mixed-frequency data based on hybrid RR-MIDAS and CNN-LSTM models. *Mathematics* 12(10):1538
- Maverick J (2024) How big is the derivatives market? Reviewed by Thomas J. Catalano, fact-checked by Vikki Velasquez, Investopedia website
- Meddahi N (2003) ARMA representation of integrated and realized variances. *Econom J* 6(2):335–356
- Mills TC (2015) Modelling stationary time series: the ARMA approach. *Time Series Econometrics: A Concise Introduction*. Palgrave Macmillan, London, pp 5–40
- Müller UA, Dacorogna MM, Davé RD, Olsen RB, Pictet OV, Von Weizsäcker JE (1997) Volatilities of different time resolutions-analyzing the dynamics of market components. *J Empir Finance* 4(2–3):213–239
- Patton AJ, Sheppard K (2015) Good volatility, bad volatility: signed jumps and the persistence of volatility. *Rev Econ Stat* 97(3):683–697
- Pong S, Shackleton MB, Taylor SJ, Xu X (2004) Forecasting currency volatility: a comparison of implied volatilities and AR(FI)MA models. *J Bank Finance* 28(10):2541–2563
- Ramos-Pérez E, Alonso-González PJ, Núñez-Velázquez JJ (2021) Multi-transformer: a new neural network-based architecture for forecasting S & P volatility. *Mathematics* 9(15):1794
- Reisenhofer R, Bayer X, Hautsch N (2022) Harnet: a convolutional neural network for realized volatility forecasting. *arXiv preprint arXiv:2205.07719*
- Richter M C, Sørensen C (2002) Stochastic volatility and seasonality in commodity futures and options: the case of soybeans. Available at SSRN 301994
- Samuelson PA (1965) Rational theory of warrant pricing. In: Henry P. McKean Jr. *Selecta*. Springer International Publishing, Cham, pp 195–232
- Şanlı S, Balçılar M, Özmen M (2025) Predicting the volatility of bitcoin returns based on kernel regression. *Ann Oper Res* 352(3):505–542
- Schwarz G (1978) Estimating the dimension of a model. *Ann Stat* 6(2):461–464
- Souček M, Todorova N (2013) Realized volatility transmission between crude oil and equity futures markets: a multivariate HAR approach. *Energy Econ* 40:586–597
- Souto HG, Moradi A (2023) Forecasting realized volatility through financial turbulence and neural networks. *Econ Bus Rev* 9(2):133–159
- Teller A, Pigorsch U, Pigorsch C (2022) Short-to long-term realized volatility forecasting using extreme gradient boosting. Available at SSRN 4267541
- Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, Kaiser Ł, Polosukhin I (2017) Attention is all you need. *Advances in neural information processing systems*, 30
- Vidal A, Kristjanpoller W (2020) Gold volatility prediction using a CNN-LSTM approach. *Expert Syst Appl* 157:113481
- Wang Y, Guo Y (2020) Forecasting method of stock market volatility in time series data based on mixed model of ARIMA and XGBoost. *China Commun* 17(3):205–221
- Wang Y, Ma F, Wei Y, Wu C (2016) Forecasting realized volatility in a changing world: a dynamic model averaging approach. *J Bank Finance* 64:136–149
- Wei Y, Liang C, Li Y, Zhang X, Wei G (2020) Can CBOE gold and silver implied volatility help to forecast gold futures volatility in china? Evidence based on HAR and ridge regression models. *Financ Res Lett* 35:101287
- Wong ZY, Chin WC, Tan SH (2016) Daily value-at-risk modeling and forecast evaluation: the realized volatility approach. *J Finance Data Sci* 2(3):171–187
- Wu Y, Wang Q (2021) LightGBM based optiver realized volatility prediction. In: 2021 IEEE international conference on computer science, artificial intelligence and electronic engineering (CSAIEE), pp 227–230
- Xiao C, Huang W, Newton DP (2024) Predicting expected idiosyncratic volatility: empirical evidence from ARFIMA, HAR, and EGARCH models. *Rev Quant Finance Account* 63(3):979–1006
- Yang K, Chen L, Tian F (2015) Realized volatility forecast of stock index under structural breaks. *J Forecast* 34(1):57–82
- Zerveas G, Jayaraman S, Patel D, Bhamidipaty A, Eickhoff C (2021) A transformer-based framework for multivariate time series representation learning. In: Proceedings of the 27th ACM SIGKDD conference on knowledge discovery and data mining, pp 2114–2124
- Zhang X (2022) A model combining LightGBM and neural network for high-frequency realized volatility forecasting. In: 2022 7th international conference on financial innovation and economic development (ICFIED 2022). Atlantis Press, pp 2906–2912

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.